

ME 300 Thermodynamics – Summer 2023

Exam 1

(Part I) _____ (30 pts)

(Part II) _____ (40 pts)

(Part III) _____ (30 pts)

TOTAL _____

Solutions

Directions

1. Open the online answers portion of the exam and download the exam (this document).
2. Make sure to show all your work!
 - a. *If you are using a tablet*, solve the problems directly on this document, showing all your work and writing as clearly as possible.
 - b. *If you are writing on paper*, solve the problems on a separate piece of paper (you do not have to print the exam), showing all your work and writing as clearly as possible.
3. Questions that require answers to be submitted to the online answers page on Canvas are written in *red italics* below as a way to remind you of what should be entered. All sketches and plots should be clearly drawn on your working paper and submitted with the upload.
4. If you have questions, please use the private chat function in Zoom to send a message directly to Dr. O'Connor so as not to disturb the other students. **DO NOT UNMUTE AND ASK YOUR QUESTION ALOUD** – this will be distracting to the other students.
5. Once the answers portion of the exam closes, scan or take very nice pictures of your work and upload it to the upload portion of the exam in a PDF document. I trust that you will not change anything on your exam after the answers portion of the exam closes. I can't enforce this – I'm really just relying on you to do the right thing. Please keep to the honor code.
6. The timeline for the exam will be the following (all times in EDT, Thursday, June 23):
 - a. Exam opens (answers portion and upload portion): 6:00 PM
 - b. Answers portion of the exam closes: 7:30 PM
 - c. Upload portion of the exam closes: 7:45 PM
7. You can anonymously report any incidents of cheating using the following Google form:
<https://forms.gle/E4wFjDtvNuginVkp9>

Academic Integrity Pledge

I pledge that I am adhering to Penn State's Code of Conduct when taking this exam.

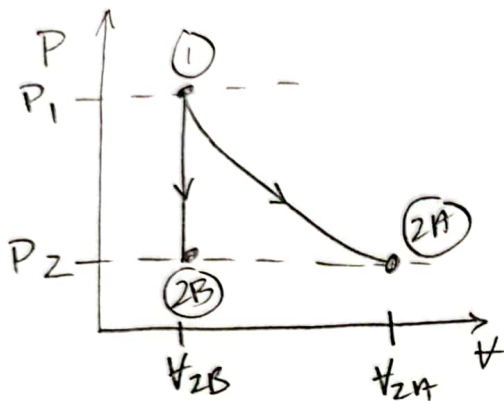
- I will not communicate with anyone during the exam except Dr. O'Connor.
- I will use only the allowed materials (calculator, notes, tables, and book), I will only access the internet to download the exam, view the e-book and notes on Canvas (no videos), and to submit my answers and upload the exam.
- I will be the author of my own work and not use any generative AI (like ChatGPT) or exam services.
- I will not change any of my written answers after the exam but before I upload my work.
- I will not cheat in any way.

It is important to me to be a person of integrity and that means that all answers on this exam are my individual answers. *In the online answers, please type your full legal name to agree to the academic integrity pledge.*

Part I: Concept Questions

1. (10 points) Two systems filled with the same mass of ideal gas start at the same state and then undergo two different processes from P_1 to P_2 , where $P_2 < P_1$: an isothermal process (Process A) and an isochoric process (Process B). Answer the following questions about these two processes:

- (3 points) Which process results in a higher volume at State 2? *Answer the multiple-choice question in Canvas.*
- (3 points) Which process results in a higher change in internal energy from State 1 to State 2? *Answer the multiple-choice question in Canvas.*
- (4 points) Explain your answer to the previous question, "Which process results in a higher change in internal energy from State 1 to State 2?" *Type your answer in the box in Canvas.*



a) $v_{2A} > v_{2B}$ +3

b) $\Delta U_A = 0$ b/c $\Delta T = 0$ +4

$\Delta U_B > \Delta U_A$ +3

c) see above

2. (10 points) Two systems of water, A and B, are at the same temperature but have different qualities: $x_A = 0.2$ and $x_B = 0.8$. Answer the following questions about the two fluids.

- (2 points) Which system has a greater mass of vapor? *Answer the multiple-choice question in Canvas.*
- (3 points) Explain your answer to the previous question. *Type your answer in the box in Canvas.*
- (2 points) Which system has the greater internal energy? *Answer the multiple-choice question in Canvas.*
- (3 points) Explain your answer to the previous question. *Type your answer in the box in Canvas.*

$$x = \frac{M_{\text{vap}}}{M_{\text{liq}} + M_{\text{vap}}} \rightarrow$$

a) system B +2

b) definition of quality +3

c) system B +2

d) $u = xu_g + (1-x)u_f$ where $u_g > u_f$
so $u_B > u_A$ +3

3. (10 points) A fluid in a piston-cylinder system expands adiabatically from volume V_1 to V_2 , where $V_2 > V_1$. Answer the following questions about this process:

- a) (2 points) What is the sign of work for this process? *Answer the multiple-choice question in Canvas.*
- b) (2 points) Is the work done on the fluid or by the fluid? *Answer the multiple-choice question in Canvas.*
- c) (2 points) What is the sign of the change in internal energy of the fluid in this process? *Answer the multiple-choice question in Canvas.*
- d) (4 points) Explain your answer to the previous question. Use an equation to explain your answer. *Type your answer in the box in Canvas.*

a) ${}_1\bar{W}_2 = \int_{V_1}^{V_2} P dV$ if $V_2 > V_1$, then ${}_1\bar{W}_2 > 0$ +2

b) work done by fluid because it expands and pushed against the surroundings +2

c) $\Delta \bar{U} = {}_1\bar{Q}_2 - {}_1\bar{W}_2 \rightarrow$ since ${}_1\bar{W}_2 > 0$

d) $\Delta \bar{U} = -{}_1\bar{W}_2$ +3
then $\Delta \bar{U} < 0$ +2

Part II: Short Problems

1. (20 points) The following table gives a read-out from NIST for an unknown species. The molecular weight of this substance is 34.08 kg/kmol. Using this information, answer the following questions:

- (4 points) Is this species an ideal gas or not at the range of pressures and temperatures shown in the table?
Answer the multiple-choice question in Canvas.
- (8 points) Use an equation of state to explain why this species is an ideal gas or not. *Type your answer in the box in Canvas.*
- (8 points) Use the critical temperature and pressure to explain why this species is an ideal gas or not. *Type your answer in the box in Canvas.*

Temperature (K)	Pressure (MPa)	Density (kg/m ³)	Volume (m ³ /kg)	Internal Energy (kJ/kg)	Enthalpy (kJ/kg)	Entropy (J/g·K)	C _v (J/g·K)
300.00	0.50000	7.1019	0.14081	557.47	627.87	2.5101	0.78245
350.00	0.50000	5.9938	0.16684	597.55	680.97	2.6739	0.78900
400.00	0.50000	5.2013	0.19226	637.92	734.05	2.8156	0.80574
450.00	0.50000	4.6007	0.21736	679.09	787.77	2.9421	0.82714
500.00	0.50000	4.1277	0.24227	721.30	842.43	3.0573	0.85095

Additional fluid properties

Critical temperature (T _c)	373.1 K
Critical pressure (P _c)	9.0000 MPa

a) ideal gas +4

$$b) P_v = RT \xrightarrow{+4} R = \frac{R_u}{M} = \frac{8314.5 \frac{\text{J}}{\text{kmol}}}{34.08 \frac{\text{kg}}{\text{kmol}}} = 243.97 \frac{\text{J}}{\text{kg} \cdot \text{K}} \xrightarrow{+1}$$

$$\begin{aligned} \rightarrow \text{State 1: } Z &= \frac{P_v}{RT} = \frac{(0.5 \times 10^6)(0.14081)}{(243.97)(300)} = 0.962 \\ \text{State 5: } Z &= \frac{P_v}{RT} = \frac{(0.5 \times 10^6)(0.24227)}{(243.97)(500)} = 0.993 \end{aligned} \quad \left. \begin{array}{l} \\ \\ \end{array} \right\} \begin{array}{l} +2 \\ \text{ideal} \\ \text{gas} \end{array}$$

$$\begin{aligned} \text{OR } \rightarrow \Delta u &= C_v \Delta T \rightarrow u_2 - u_1 = (597.55 - 557.47) = 40.08 \text{ kJ/kg} \\ C_v \Delta T &= 785.5(50) = 39.27 \text{ kJ/kg} \end{aligned} \quad \left. \begin{array}{l} \\ \uparrow \\ \text{average} \end{array} \right\} \text{close}$$

c) $T > T_c$
 $P < P_c$ } there are the conditions for an ideal gas
because $V_{\text{mol}} \approx V_{\text{b/w mol}}$ at these low pressures.
+4

2. (20 points) Water begins at State 1 in a saturated liquid state at 2 MPa and is heated in an isobaric process until a temperature of 600 K at State 2.

- (2 points) Write the assumptions necessary to solve for State 2 and the heat and work. *Enter these in the Canvas quiz – you do not need to also write them on the page.*
- (2 points) What is the temperature of the water at State 1 in Kelvin [K]? *Show your work below or on a separate page and enter this value in the Canvas quiz.*
- (2 points) What is the phase of the water at State 2? *Show your work below or on a separate page and answer this question in the Canvas quiz.*
- ⁸ (6 points) What is the work done during this process in kilojoules [kJ] if the mass inside the cylinder is 1 kg? *Show your work below or on a separate page and enter this value in the Canvas quiz.*
- ⁶ (8 points) What is the heat added to the system from State 1 to State 2 in kilojoules [kJ] if the mass inside the cylinder is 1 kg? *Show your work below or on a separate page and enter this value in the Canvas quiz.*

a) S.C.S. +1
 quasi-eq process +1
 $\Delta KE = \Delta PE = 0$

b) Given: $x_1 = 0$
 $P_1 = 2 \text{ MPa}$ $\rightarrow$ $T_1 = T_{\text{sat}} = 485.53 \text{ K}$ ⁺² on Table D.2

c) Given: $P_2 = P_1 = 2 \text{ MPa}$
 $T_2 = 600 \text{ K}$ $\rightarrow T_2 > T_{\text{sat}} @ 2 \text{ MPa} \rightarrow$ vapor ⁺²

d) ${}_1\bar{W}_2 = \int_{v_1}^{v_2} P dv$ ⁺² $= M \int_{v_1}^{v_2} P dv$ ⁺² $= MP(v_2 - v_1)$ because isobaric

$\rightarrow v_1 = v_f = 0.0011767 \text{ m}^3/\text{kg}$ on Table D.2 ⁺¹

$\rightarrow v_2 = 0.13262 \text{ m}^3/\text{kg}$ on Table D.3 ⁺¹

${}_1\bar{W}_2 = (1)(2 \times 10^6)(0.13262 - 0.0011767) \rightarrow$ ${}_1\bar{W}_2 = 262.887 \text{ kJ}$ ⁺²

$$c) \Delta \bar{U} = {}_1\bar{Q}_2 - {}_1\bar{W}_2 \quad +2$$

$${}_1\bar{Q}_2 = M \Delta u + {}_1\bar{W}_2 \rightarrow u_1 = u_f = 906.14 \text{ kJ/kg} \quad +1 \text{ from Table D2}$$

$$\rightarrow u_2 = 2820.4 \text{ kJ/kg} \quad +1 \text{ from Table D3K}$$

$${}_1\bar{Q}_2 = (1)(2820.4 - 906.14) + 262.887$$

$$\boxed{{}_1\bar{Q}_2 = 2177.15 \text{ kJ}} \quad +2$$

Part III: Analysis (30 points)

The story: Argon undergoes a four-step cycle to produce power for a small factory. The argon is sealed in a piston-cylinder device with a mass of 10 kg. Assume the following properties for argon ($R=208 \text{ J/kg-K}$ and $c_v=312 \text{ J/kg-K}$) and use the cycle below to answer the questions.

- Process 1-2: Isothermal compression
- Process 2-3: Isochoric heating
- Process 3-4: Isothermal expansion
- Process 4-1: Isochoric cooling

- a) (4 points) List the assumptions you'll need to solve for the state points as well as the first law quantities (heat, work, change in internal energy). *Enter these in the Canvas quiz – you do not need to also write them on the page.*
- b) (12 points) Use the state points below to solve for the remainder of the states (P in [kPa], v in [m^3/kg], and T in [K]). *Show your work below or on a separate page and enter this value in the Canvas quiz.*

State	Pressure [MPa]	Volume [m^3/kg]	Temperature [K]
1	0.15	4.853	350
2	3	0.2427	350
3	6.857	0.2427	800
4	0.3429	4.853	800

- c) (7 points) What is the net work of the cycle? *Show your work below or on a separate page and enter this value in the Canvas quiz.*
- d) (8 points) Draw this cycle on a P-V diagram, labeling all the state points. *Show your work below or on a separate page – you do not need to submit anything on the Canvas quiz for this part.*

a) ideal gas +1

$G = \text{const} +1$

quasi-cf processes +1

$\Delta KE = \Delta PE = 0 +1$

Given: $M = 10 \text{ kg}$

$R = 208 \text{ J/kg-K}$

$C_v = 312 \text{ J/kg-K}$

b) State 1: $v_1 = \frac{MRT_1}{P_1} = \frac{(10)(208)(350)}{0.15 \times 10^6} \rightarrow \boxed{v_1 = 4.853 \text{ m}^3} +1$

State 2: $\boxed{T_2 = T_1 = 350 \text{ K}} +1$

$v_2 = \frac{MRT_2}{P_2} = \frac{(10)(208)(350)}{3 \times 10^6} \Rightarrow \boxed{v_2 = 0.2427 \text{ m}^3} +1$

State 3: $V_3 = V_2 = 0.2427 \text{ m}^3$ $+1$

$$P_3 = \frac{MRT_3}{V_3} = \frac{(10)(208)(800)}{0.2427} \Rightarrow P_3 = 6.857 \text{ MPa} \quad +1$$

State 4: $T_4 = T_3 = 800 \text{ K}$ $+1$

$$V_4 = V_1 = 4.853 \text{ m}^3 \quad +1$$

$$P_4 = \frac{MRT_4}{V_4} = \frac{(10)(208)(800)}{4.853} \Rightarrow P_4 = 0.3429 \text{ MPa} \quad +1$$

c) $\bar{W}_{\text{net}} = \bar{W}_2 + \bar{W}_3 + \bar{W}_4 + \bar{W}_1$
 $\bar{W}_3 \rightarrow 0$ $+1/2$ b/c isochoric

$$\bar{W}_2 = \int_{V_1}^{V_2} P dV \quad \text{for ideal gas } P = \frac{MRT}{V} \Rightarrow \bar{W}_2 = \int_{V_1}^{V_2} \frac{MRT}{V} dV = MRT \ln(V_2/V_1)$$

$$= (10)(208)(350) \ln\left(\frac{0.2427}{4.853}\right)$$

$$\bar{W}_2 = -2180.743 \text{ kJ} \quad +1$$

$$\bar{W}_4 = MRT_3 \ln(V_4/V_3) = (10)(208)(800) \ln\left(\frac{4.853}{0.2427}\right) \Rightarrow \bar{W}_4 = 4984.56 \text{ kJ}$$

$$\bar{W}_{\text{net}} = 2803.81 \text{ kJ} \quad +1$$

